

Portfolio Choice in the Presence of Housing

(Cocco, 2005, Review of Financial Studies)

A Replication by Zachary Orlando *

Introduction

Why do so few individuals own stock early in life? As labor income is relatively uncorrelated with stock market returns (Cocco et al., 2005), young individuals, who hold a great amount of future labor income, should invest as if they are already significantly exposed to safe assets. We would then expect them to allocate close to all of their investments into risky assets like stocks. Yet in the data, only 26% of under 35 adults in the 1989 PSID reported owning any stock at all.

“Portfolio Choice in the Presence of Housing” shows that individual’s housing investment decisions play an important role in explaining this puzzle. Housing risk crowds out stock market holdings. Moreover, as most housing purchases occur while young, this reduces the magnitude of liquid assets and depletes the incentive to invest in stock consistent with relatively low assumed “participation costs”. Overall, stock market participation when factoring in housing, is predicted to rise over the lifecycle consistent with the data.

This replication has two parts:

- In Part I, I detail the lifecycle portfolio-choice model of the paper.
- In Part II, I then replicate its main exhibits.

I wanted to be up front on some results that I presently cannot replicate:

- My simulation predicts far higher stock market participation rates than reported, particularly early in life. This has been robust across all the potential model / solution-method versions I tried ¹.
- I also found that house price risk, as it is parameterized to be perfectly correlated with persistent income risk, instead somewhat *crowds-in* stock market investment as a hedge.

In Appendix A I summarize my thoughts on these differences, show that my solution method replicates the closely related paper (Cocco et al., 2005) and compare / contrast the properties of my simulated model with those reported in the paper.

*UW Madison, zorlando@wisc.edu. Code is available on GitHub.

¹A subset of reasonable parameterizations are detailed in Appendix E.

Part I: The Model

Consider the partial-equilibrium problem of an investor who receives uninsurable labor market earnings for T (10, every model-period represents five years) periods and must decide how much to invest in the stock market versus bonds and additionally how large of a home to own.

The dynamic problem can be recursively represented as:

$$V_t(\text{State}_t) = \max_{C_t, H_t, D_t, FC_t, \alpha_t} \left[\frac{(C_t^{1-\theta} H_t^\theta)^{1-\gamma}}{1-\gamma} + \beta E_t[V_{t+1}(\text{State}_{t+1})] \right], \quad \forall t \leq T.$$

Where C_t is consumption, H_t is housing today, D_t is mortgage debt held today, FC_t is whether the investor pays the fixed stock market entry fee at period t, α_t is the investor's stock market holdings today.

Note in this model, housing is a **continuous** variable and investors are forced to hold a positive minimum amount of housing.

State_t consists of “Cash-on-Hand” (“Cash”), X_t , housing chosen yesterday, H_{t-1} , whether the investor is already a stock market participant IFC_t , an “involuntary movement” state Inv_Move_t and the aggregate labor market state today η_t .

Housing depreciates at a rate δ (1% per annum), there is a one-time stock market participation fee F (\$1000) and a house-sale adjustment cost fee of $\lambda(8)\%$.

In each period, the investor's budget constraint is:

$$S_t + B_t = \begin{cases} X_t - C_t - FC_t F - \delta P_t H_{t-1} + D_t, & \forall t \text{ s.t. No House Trade,} \\ X_t - C_t - FC_t F - \delta P_t H_{t-1} + D_t + (1 - \lambda)P_t H_{t-1} - P_t H_t, & \forall t \text{ s.t. House Trade.} \end{cases}$$

S_t is the total value of their stock market holdings in period t, B_t is the value of their holdings of a risk-free bond and P_t is the house price. The model assumes stocks and bonds must obey short-sale constraints: $S_t \geq 0, B_t \geq 0$.

In any period, an investor's “Liquid Wealth” refers to their total earnings from their portfolio (net of debt) last period:

$$LW_t = R_t S_{t-1} + R^F B_{t-1} - R^D D_{t-1}$$

Here, the risk-free rate is 2%, the mortgage rate is 4% and the stock market return is a random variable (with annual mean 10%). “Cash” in any period is the sum of liquid wealth and labor market earnings:

$$X_t = LW_t + Y_t$$

The investor additionally values their bequest at T + 1,

$$W_{T+1} = X_{T+1} - \delta H_T P_{T+1} + (1 - \lambda) H_T P_{T+1}$$

as if it was another period of life:

$$U_1 = E_1 \left[\sum_{t=1}^T \beta^{t-1} \frac{(C_t^{1-\theta} H_t^\theta)^{1-\gamma}}{1-\gamma} + \beta^T \frac{W_{T+1}^{1-\gamma}}{1-\gamma} \right].$$

Investors face three main sources of uncertainty:

1. Labor Income risk
2. Housing risk
3. Stock Market risk

Labor income

An investor at age t 's labor income (in logs) follows an exogenous process given by:

$$\tilde{y}_{it} = \begin{cases} f(t, Z_{it}) + \tilde{u}_{it}, & \text{for } t \leq K, \quad (\text{Working}) \\ f(t, Z_{it}), & \text{for } t > K. \quad (\text{Retirement}) \end{cases}$$

where K is the date of retirement (here, 65 years). $f(t, Z_{it})$ is a deterministic income process and risk, $\tilde{u}_{it}$, is given by the sum of a persistent aggregate process and an idiosyncratic transitory shock.

$$\tilde{u}_{it} = \eta_t + \omega_{i,t}$$

$$\eta_t = \phi\eta_{t-1} + \epsilon_t$$

$\omega_{i,t} \sim N(0, \sigma_\omega^2)$, $\epsilon_t \sim N(0, \sigma_\epsilon^2)$. After retirement, individuals receive a constant pension which is determined by their education group.

Housing and Debt

The baseline model only considers the problem of homeowners. All investors must purchase at least $H \geq H_{min}$ (\$20,000 in 1992 dollars) worth of housing in the first period and retain at least this amount their entire lives.

House prices follow a log-linear time trend with a residual component:

$$\log(P_t) = p_t = bt + p'_t$$

To simplify computation, Cocco imposes that p'_t , the risk in house prices, is perfectly correlated with the aggregate labor income state η_t .

An investor may also be hit with an “Involuntary Movement” shock with probability π that forces them to sell their home. Because selling housing is subject to a proportional adjustment cost λ , individuals lose resources when forced to sell when they otherwise would not have chosen to do so.

Households can costlessly adjust their mortgage debt but must respect a down-payment constraint:

$$D_t \leq (1 - d)P_t H_t$$

where d is the minimum down-payment (15%). This can be equivalently expressed as a Loan - to - (Home) Value constraint:

$$LTV \leq (1 - d)$$

Stock Market

Stock market returns are log-normally distributed:

$$\log(R_t) = \mu + \iota_t$$

$$\iota_t \sim N(0, \sigma_\iota^2).$$

Table 1. (Annual) ² Parameters used for the replication.

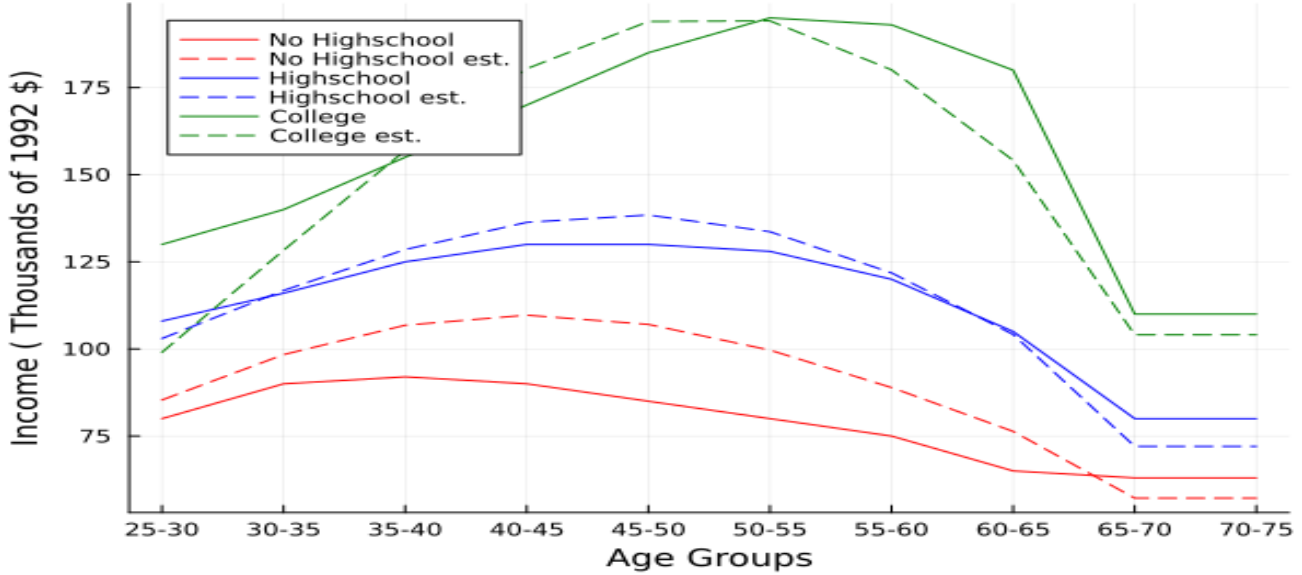
Description	Parameter	Value	Description	Parameter	Value
Risk aversion	γ	5	House-price growth	b	1%
Discount factor	β	0.96	SD of house prices	$\sigma_{\hat{p}}$	0.062
Preference for housing	θ	0.10	Aggregate persistence	ϕ	0.748
Down payment	d	0.15	<i>SD idiosyncratic income shocks</i>		
Depreciation rate	δ	0.01	No high-school degree	$\sigma_{\omega,1}$	0.136
Involuntary move	π	0.032	High-school degree	$\sigma_{\omega,2}$	0.131
Transaction cost	λ	0.08	College	$\sigma_{\omega,3}$	0.133
Riskless rate	$\hat{R}_F - 1$	0.02	SD aggregate income shocks	σ_{η}	0.019
Mortgage rate	$R_D - 1$	0.04			
Mean stock return	$\exp(\mu + \sigma_{\eta}^2/2) - 1$	0.10			
Std. of log stock return	σ_{η}	0.1674			
One-time entry cost	F	\$1000			
Minimum House Size	H_{min}	\$20000			

(a) Preferences and stock returns

(b) Housing and income-process parameters

Table 1: Parameters used in the model
(All parameters used in the model are directly from the paper.)

Figure 1: Deterministic Income Profile



Solid lines are reported by the paper (I had to approximate the values from Figure 1. of the paper) and fed into the model exercise. Dashed lines are my own estimates from the PSID.

²Panel b) is taken from a different table where the time-period is unspecified. The housing values are annual but it is possible the income process parameters are five-year values. I cover this possibility in Appendix E. Appendix C covers how I scaled these values into five-year parameters.

Part II: Model Replication

It is unclear how the labor and housing market process parameters were scaled to their five-year levels in the original paper. My baseline ³. estimates are that these reported parameters are annual and are scaled using:

- The variance of the housing process does not change from one to five years (this is equivalent to assuming households make housing choices once every five years).
- Five-year persistence is ϕ^5 (all labor income process parameters were scaled using this same method as for instance (Fonseca et al., 2024)).
- The variance of the transitory and persistent processes does not change. In conjunction with the scaling of ϕ this is the same as scaling the persistent processes' innovation variance up by $\frac{(1-\phi^{10})}{(1-\phi^2)} \sim 2.15$. This is equivalent to assuming investors experience one shock every five years.
- The regression coefficient of persistent shocks on house price shocks, κ_η , is given by the ratio of their respective five-year standard deviations.

The details of the approach I used to solve and simulate the model, and how they compare to the paper's method, are contained in Appendix B. To the best of my ability, I am using the same methods as those detailed in the paper, with exception of using linear interpolation instead of cubic spline.

Replicated Results

The four replicated exhibits are:

1. Portfolios as a share of assets, split by financial net worth. This compares the behavior of broadly "poor" to "rich" investors.
2. Portfolios as a share of assets, split by age-group. This compares the behavior of investors across the life cycle.
3. Tobit regressions of stock shares of assets against financial and demographic determinants (holding the path of aggregate shocks fixed over time).
4. Portfolios as a share of financial assets, comparing a model with house price risk to one with no risk. This allows us to learn about the crowding-out effect of house price risk on stock market participation.

Following the paper, I construct Tables 3 and 4 by simulating a sample of consumption/portfolio choices by age and education in the same proportion as the age-education distribution in the 1989 PSID ⁴. As the deterministic profile is constructed using age-group dummies controlling for marital status and number of additional members of the household, this exercise is for unmarried investors with no dependents.

Like in the paper, I allow the simulated households to differ in their history of both *the aggregate shock* and the other shocks. This provides the unconditional portfolio allocations generated by the model.

³My appendix F reports a matrix of alternative scaling approaches I also considered, and that using them did not affect the final result)

⁴The weights I used were taken from the appear and are shown in Appendix D.

By examining Table 3, we see in the original paper there is a stark distinction between rich and poor investors. Rich investors almost entirely invest in stocks (as opposed to bonds) as predicted by standard portfolio choice models. However, poor investors invest much more in bills, as their portfolio is disproportionately dominated by real estate.

In my replication, the pattern is still present, but the difference between rich and poor is much less stark. My simulation predicts the stock share of poor investors to be around 95% as compared to 25% in the paper (this number appears to be relatively sensitive to the solution method, a previous version of my notes predicted poor investors hold 79% in stocks). My replication predicts that rich investors hold practically 100% in stock, compared to 95% in the original model.

In general, while the predicted values are relatively close for rich investors, my replication consistently predicts meaningfully higher amounts of stock market participation for poorer investors.

Furthermore, comparing the mean debt share to the mean real estate share of financial assets, we can see that in my simulation they are in a ratio of 0.85:1, while in the the paper their ratio is 0.55:1. As 85% is the loan-to-value limit, this suggests more poor investors are *borrowing constrained* in my simulation than in the reported model.

Table 3: Portfolio shares of assets by financial net worth predicted by the model
(author’s estimates in **bold** parentheses)

Asset	Liquid assets		Financial assets		Total assets	
	< 100k	≥ 100k	< 100k	≥ 100k	< 100k	≥ 100k
Stocks	0.252 (0.953)	0.952 (0.994)	0.049 (0.281)	0.460 (0.569)	0.017 (0.084)	0.213 (0.354)
Bonds (Bills)	0.748 (0.047)	0.048 (0.006)	0.029 (0.001)	0.013 (0.004)	0.008 (0.000)	0.006 (0.003)
Liquid assets	1.000	1.000	0.078 (0.281)	0.473 (0.573)	0.025 (0.082)	0.219 (0.366)
Real estate			0.922 (0.719)	0.527 (0.427)	0.248 (0.161)	0.251 (0.286)
Financial assets			1.000	1.000	0.273 (0.245)	0.470 (0.653)
Human capital					0.727 (0.755)	0.530 (0.357)
Total assets					1.000	1.000
Debt			0.509 (0.596)	0.034 (0.142)	0.141 (0.124)	0.018 (0.070)
Stock mkt. part.	0.253 (0.919)	0.974 (1.000)				

Notes: The first line in each cell is the share reported in (Cocco, 2005).

The value in bold parentheses underneath is the author’s own estimate obtained from simulation.

“Liquid assets” refers to the sum of stocks and bonds. “Financial assets” are liquid assets plus real estate value. Total assets includes “Human Capital” - constructed as discounted expected future labor income using an annual discount rate of 5%. I approximated this using monte-carlo simulation.

“Financial Net Worth”, is computed by subtracting debt from each asset’s definition.

Table 4. provides average portfolios predicted by the model across the lifecycle. While stock market shares are still upward sloping over the lifetime, we can again see much higher stock market participation in my simulated data. On the extensive margin, we see that 81% of young households participate in the stock market, as opposed to 2% reported in the paper. On the intensive margin, we see that stocks represent a meaningful share of total lifetime assets for all age-groups while in the paper they represent only a small share of investment.

I also noticed that debt, as a proportion of financial assets, is relatively flat under the reported model whereas it falls in my model. To me, this suggests that agents are using debt to consumption smooth and potentially “lever up” at old age in the reported model, whereas under my simulation, debt is much more associated with the initial house purchase. Comparing the debt to real estate shares of financial assets, we can again see evidence that more young investors are borrowing constrained in my simulation than reported in the paper.

For instance, agents under 35 have a mean portfolio share in real estate of 85% and a debt share of 72% in my simulation. Debt to real estate value is again in a ratio of 0.85:1, the maximum LTV ratio. The corresponding number in the paper is 0.7, suggesting some young investors are not borrowing the maximum possible.

Table 4: Portfolio shares by age predicted by the model
(author’s estimates in **bold** parentheses)

Asset	Liquid assets				Financial assets				Total assets			
	<35	35–50	50–65	≥65	<35	35–50	50–65	≥65	<35	35–50	50–65	≥65
Stocks	0.02 (0.87)	0.21 (0.99)	0.61 (1.00)	0.72 (0.98)	0.01 (0.15)	0.06 (0.35)	0.24 (0.60)	0.10 (0.45)	0.00 (0.03)	0.01 (0.10)	0.10 (0.33)	0.07 (0.38)
Bills	0.98 (0.13)	0.79 (0.00)	0.39 (0.00)	0.28 (0.02)	0.03 (0.00)	0.02 (0.00)	0.04 (0.00)	0.01 (0.01)	0.00 (0.00)	0.01 (0.00)	0.02 (0.00)	0.01 (0.01)
Liquid assets	1.00	1.00	1.00	1.00	0.03 (0.15)	0.08 (0.35)	0.28 (0.60)	0.11 (0.46)	0.00 (0.03)	0.02 (0.10)	0.12 (0.32)	0.08 (0.40)
Real estate					0.97 (0.85)	0.92 (0.65)	0.72 (0.40)	0.89 (0.54)	0.13 (0.13)	0.15 (0.15)	0.29 (0.21)	0.72 (0.47)
Financial assets					1.00	1.00	1.00	1.00	0.13 (0.15)	0.17 (0.25)	0.41 (0.53)	0.80 (0.87)
Human capital									0.87 (0.84)	0.83 (0.75)	0.59 (0.46)	0.21 (0.14)
Total assets									1.00	1.00	1.00	1.00
Debt					0.68 (0.72)	0.43 (0.55)	0.27 (0.20)	0.51 (0.05)	0.09 (0.11)	0.07 (0.13)	0.12 (0.10)	0.42 (0.04)
Stock mkt. part.	0.02 (0.81)	0.21 (0.97)	0.62 (1.00)	0.73 (1.00)								

Notes: First line in each cell is the model share reported by (Cocco, 2005). The value in bold parentheses underneath is the author’s own estimate obtained from simulation.

Table 5 shows the determinants of stock market portfolios shares and the absolute investment in stocks. It conditions instead on a single aggregate shock history that best matches the path that actually occurred. I generated this by inspecting the path of the house price index process against a linear time trend from 1970 - 1989 ⁵ .

All households of the same age will have experienced the same history of shocks across their lifetimes. However, households of different ages will not have experienced the same histories (as age and cohort diverge in this simulation). Furthermore, I follow the paper in specifying the shocks experienced by those born before 1945 (i.e. pre-1970 shocks) to the unconditional mean (of 0).

Table 5: Tobit regressions of stock holdings on investor characteristics

	Stock relative to liquid assets	Stock relative to financial assets	Stock relative to total assets	Stocks
Intercept	-8.405 (1.464)	0.423 (0.457)	0.085 (-0.048)	-94223.460 (-90766.63)
Total income	1.59e-05 (5.11e-07)	3.73e-07 (-8.17e-07)	-1.67e-07 (-1.04e-06)	0.209 (0.341)
Financial net-worth	2.92e-04 (-5.87e-07)	1.58e-06 (9.50e-07)	1.35e-06 (9.59e-07)	
Age	4.608 (-0.007)	0.021 (0.003)	0.014 (0.006)	12110.850 (2163.011)
Relative real estate	-48.834 (0.108)	-0.762 (-0.605)	-0.266 (-0.163)	
Relative mortgage	48.174 (0.291)	0.725 (0.592)	0.264 (0.183)	
Real estate				0.245 (0.431)
Mortgage				-0.957 (-0.489)

Notes: This table compares the reported results of cross-sectional Tobit regressions of stock holdings (relative to liquid, financial, and total assets) and the dollar value of stocks on investor characteristics. Values in bold parentheses underneath are my estimates. Independent variables include total income, financial net-worth, age, and measures of real estate and debt relative to financial net-worth. Real estate and mortgage variables enter the dollar regression directly. Financial net-worth is constructed by subtracting debt from financial assets. Relative real-estate and mortgage variables are constructed by dividing real estate assets and debt respectively by financial net worth.

Differences in the sign of financial net worth can be explained by differences in the magnitude of stock market participation between the reported model and my replication. This is because the underlying portfolio rule (for investing in stocks) is decreasing in financial net worth, conditional on stock market entry ⁶. However, stock market entry rises with financial net worth.

Lower participation in the paper generates a difference in sign for liquid assets. My estimate picks up the shift towards bills by the rich, while the paper's picks up stock market entry. My sense is that normalizing by financial or total assets relative to liquid assets weights the results more towards those investors who have a lower denominator. Intuitively, those who have less financial net worth (conditional on age) tend to participate less and therefore drive a positive effect.

⁵The best - match was: 1970-1974 (below trend), 1975 - 1979 (above trend), 1980-1984 (on trend) and 1985-1989 (on trend). These correspond one to one with the realization of the three grid-point aggregate shock.

⁶As financial net worth rises, the implicit holdings of safe assets through future labor income fall in significance as a share of lifetime income, and therefore investors substitute towards bills.

I replicate the positive correlation between mortgage size and stock shares reported in the paper. This is likely influenced by lifetime income: individuals with higher incomes over their lives can afford to take more risk.

Finally, I replicate the substitution between real estate shares and stock shares (except when normalizing relative to liquid assets).

In **Table 6.**, I report portfolio shares of financial assets from the baseline model (split by financial net worth) compared to the same model but solved with no house price risk.

Table 6: Portfolio shares relative to financial assets: Baseline vs. No House-Price Risk

Asset	Baseline (with house-price risk)		No house-price risk	
	< 100k	$\geq 100k$	< 100k	$\geq 100k$
Stocks	0.049 (0.281)	0.460 (0.569)	0.056 (0.273)	0.489 (0.571)
Bills	0.029 (0.001)	0.013 (0.003)	0.022 (0.000)	0.002 (0.003)
Liquid assets	0.078 (0.282)	0.473 (0.572)	0.078 (0.280)	0.491 (0.581)
Real estate	0.922 (0.718)	0.527 (0.427)	0.922 (0.726)	0.509 (0.426)
Debt	0.508 (0.596)	0.034 (0.141)	0.473 (0.601)	0.013 (0.133)
Stock mkt. part.	0.253 (0.919)	0.974 (1.000)	0.311 (0.912)	1.000 (1.000)

Notes: First line in each cell is the share reported by (Cocco, 2005). Bold values in parentheses below are the author's estimates from simulation.

We can see that in the reported results, (primarily poor) investors increase their stock shares when house price risk is shut-off. This is evidence for house price risk crowding out stock market investment.

However, in my simulation, the opposite occurs, under no house price risk, participation and stock shares actually **fall** somewhat. This is consistent with the perfect correlation of housing with income risk making stocks an attractive hedge - and actually crowding in stock market investment.

Appendix

A: Thoughts on Differences

What generates the different behavior, particularly the higher stock market participation rates, in my model compared to what is reported in the paper?

I have done the following to investigate:

1. Use my model to replicate the key policy function and simulation figures reported in the closely related paper (Cocco et al., 2005).
2. Investigated whether stock market participation in my model with housing stripped out matches the rate reported in (Cocco, 2005).
3. Repeated participation fee and minimum house size comparative statics exercises in the paper to see if similar forces are at play in my simulated model.

My conclusions are the following:

- My code replicates (Cocco et al., 2005), which confirms a discrepancy likely does not arise in the basic lifecycle portfolio choice setup, and instead appears in either the parameterization or the elements new to this model (housing, collateralized debt or the persistent income process) .
- My model predicts higher stock market participation than reported (89% vs. 76%) when housing is stripped out. This points to a discrepancy even without housing - raising the cost of stock market participation to \$5000 (as if the figure was “annualized”) closes this gap.
- There exists a second, considerable discrepancy in the housing sector in addition to whatever generates the gap in the model without housing. I think this has to do with borrowing constraints being much tighter in my model than in the paper.
 1. Almost all young and poor investors in my simulation are credit constrained, this is not present in the reported results.
 2. Unlike in the paper, stock market participation is insensitive to the minimum housing size in my model. I believe the extra access to liquid debt provided by more housing is countervailing the fall in cash balances due to a higher initial housing purchase.

Replication of Cocco, Gomes & Maenhout (2005)

First, I have replicated the closely related lifecycle portfolio-choice model of (Cocco et al., 2005). The model of (Cocco, 2005) can be turned into the former model with the following changes ⁷:

- The housing sector is stripped out.
- Stock market entry costs are set to 0.
- The persistent income process is replaced by a permanent process.
- Stock risk is normally distributed instead of log-normally distributed.
- Bequest is shut-off (in the baseline).
- A probability of death schedule (p_t) is used to compute continuation values.

⁷I thank Cutberto Frias Sarraf, who provided me his own replication code of this paper for reference, which contained deterministic income process and death probability values derived from code from Francisco Gomes.

The recursive problem of the translated model is

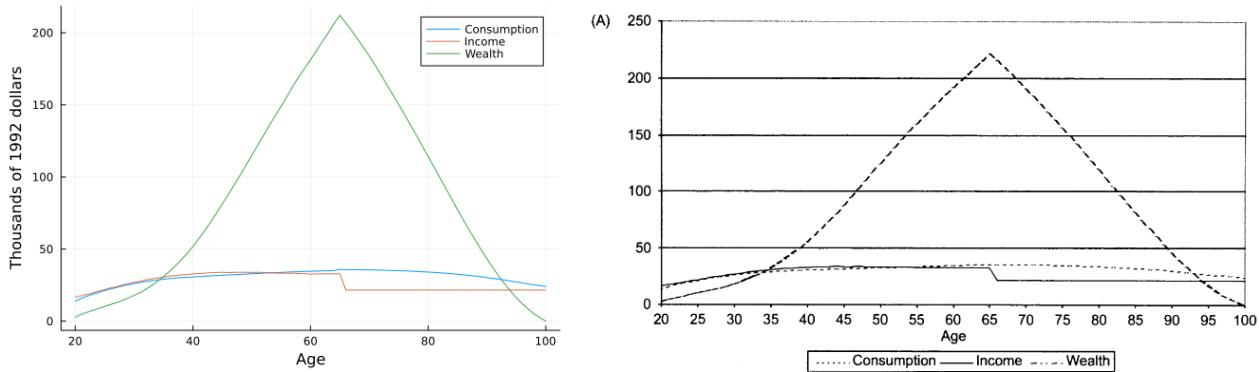
$$V_{it}(X_{it}) = \max_{\substack{C_{it} \geq 0 \\ 0 \leq \alpha_{it} \leq 1}} \left\{ \frac{(C_{it})^{1-\gamma}}{1-\gamma} + \beta p_t E_t[V_{i,t+1}(X_{i,t+1})] \right\}, \quad t < T \quad (1)$$

$$\text{s.t.} \quad X_{i,t+1} = Y_{i,t+1} + (X_{it} - C_{it})(\alpha_{it}R_{t+1} + (1 - \alpha_{it})\bar{R}_f). \quad (2)$$

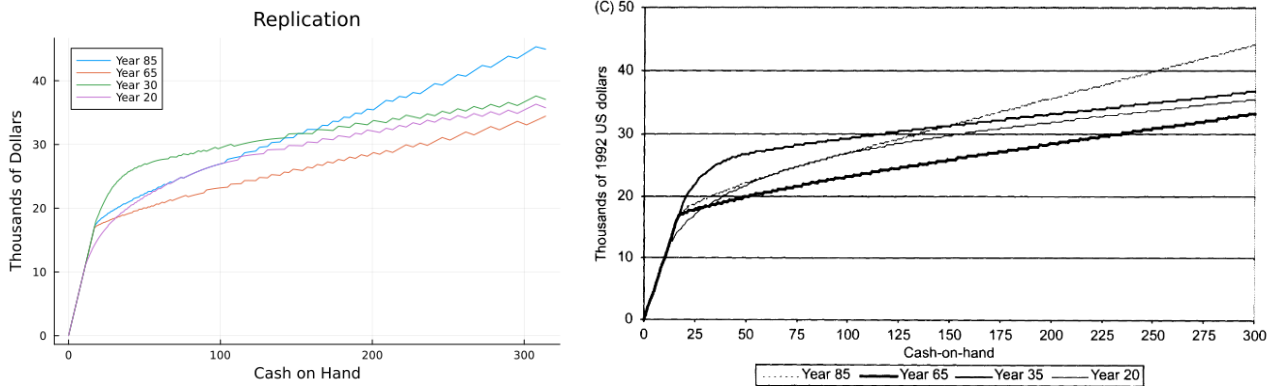
This leaves a lifecycle portfolio choice problem with only one state (as permanent income risk can be normalized away), which is cash. Furthermore, as there is no debt or discrete decisions, cash is now bounded from below by zero and cubic-spline interpolation works well.

The below figures show that after imposing these changes, my model is indeed consistent with that of (Cocco et al., 2005).

Simulated Lifecycle Means: Replication vs CGM Reported



Consumption Policy: Replication vs. CGM Reported



This suggests the difference arises in either the elements I removed and/or the parameterization of the model.

Stripping out Housing

Second, the paper checks that it is indeed housing that is driving reduced participation by reporting the participation when solving the model with housing stripped out. Participation rises from 33% in the baseline model to 76% in the model without housing. I instead that participation is flat or even *falls* when housing stripped out. I find a participation rate of

around 94% in the full model and 89% in the model without housing. This may be because borrowing is shut off in the model without housing.

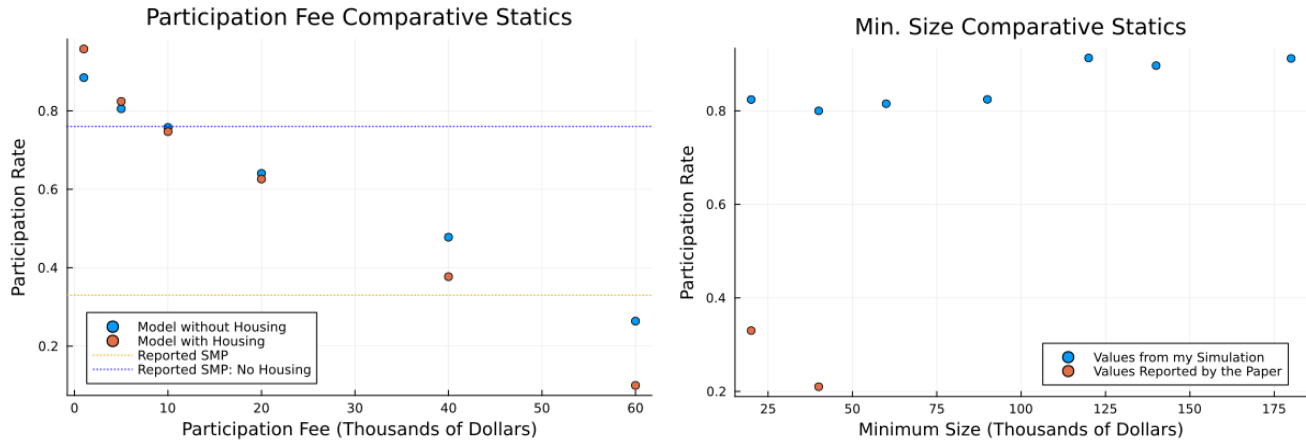
Furthermore, there is a discrepancy between my without-housing number and that reported by the paper. Because the model without housing also implies there is no involuntary movement and no debt, this is a simple extension of the (Cocco et al., 2005) paper to include persistent rather than permanent income risk and a one-time stock market participation fee. Therefore, this difference is most likely due to differences in parameters.

Having investigated the parameter space ⁸, the only change in parameters that seems to close this gap is increasing the fixed cost of participation by 5 - 10 times. This may suggest the one-time fee was “annualized” in the paper by dividing it by 5.

Comparative Statics Exercises

Third, I recorded the participation rate when solving the model under different minimum housing size and stock market participation parameters. This repeats the comparative statics exercises in the paper.

Comparative Statics: Replicated Model vs. Paper



Note that the minimum size comparative statics plots was generated using a participation fee of \$5000.

The left plot shows that to bring my model without housing in accordance with the reported value (where the blue line intersects the blue dot), I would need a participation fee of \$10,000 (10 times what is reported in the paper). To bring the model with housing into accordance with the reported overall participation rate in the paper (the orange line), I would need a participation fee of around \$40,000 (40 times what is reported in the paper).

The right plot shows that in my model, the minimum house size parameter has limited impact on stock market participation. However, in the paper increasing the minimum house size greatly reduces participation (raising H_{min} from \$20,000 to \$40,000 reduced participation from 33% to 21%). This could suggest many investors hold close to the minimum level of housing in the paper’s simulation, so the constraint affects their decision making. If this is the case, this would be very strange as Table 4. shows that real estate is the primary investment for most investors. If investors hold little real estate (in magnitude) then they simple are not saving. I cannot rationalize this behavior in the presence of a powerful warm-glow bequest.

By contrast, the median investor in my simulation owns a home of size $H = 65K$. The 99th percentile house size is 140K. Only 1% hold the minimum housing size.

However, even when the minimum housing size is \$140,000, the participation rate strangely does not fall in my simulation. I believe this is because a “collateral channel” is counteracting the effect of falling liquid balances from the initial housing purchase. When the minimum house size is increased, investors who are forced to purchase more housing will also

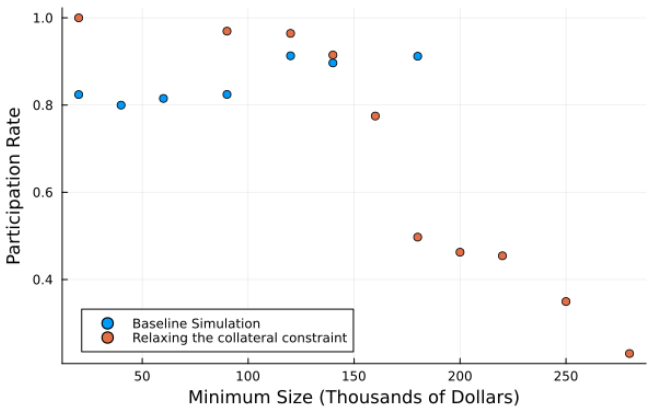
⁸I detail the combinations I have investigated in Appendix E

have looser collateral constraints. This allows them to borrow more in subsequent periods, and potentially pay the stock market participation fee.

It is plausible that the collateral channel would play a larger role in my simulation because as discussed in Part II Tables 3. and 4., it is likely more investors in my model are borrowing constrained than in the paper.

To investigate whether falling liquid balances due to increased spending on housing is being offset by rising liquid balances due to loosened collateral constraints, I repeated the second comparative statics exercises after shutting off the collateral channel. I did this by finding the maximum LTV ratio needed to fully slacken the collateral constraint (I needed to raise the maximum LTV ratio to 8.5 times - 10 times what is allowed in the paper). In the simulated model with fully slackened collateral constraints, raising the minimum house size should no longer lead to extra borrowing as no agent is borrowing-constrained.

Comparative Statics: Collateral Constraint Relaxed vs. Baseline
Min. Size Comparative Statics



Note: these participation rates are generated using a participation fee of \$5000.

When agents can freely borrow, participation rises (for the same minimum house size). For example, participation jumps from 80% to 100% at the baseline minimum size of \$20,000. In the model with unconstrained borrowing, raising the minimum house size reduces participation. This is consistent with the initial house purchase reducing liquid balances and the benefit of stocks. However, this effect only seems to change behavior for relatively high minimum house sizes, which redirect the lion's share of borrowing towards the house purchase and away from consumption-smoothing / stock-market entry.

B: Numerical Technique

Solution

Because this is a finite horizon model, I use backwards induction to approximate each education group’s value and policy functions⁹.

Following the paper, for each state I find the optimal decision using equally-spaced grid-search over stock market entry, three “share” choice variables (the share of stocks to liquid assets, the household’s loan-to-home value ratio, which implies a unique debt choice given a price level and a choice of housing, the share of maximum consumption actually consumed, whose formula varies by whether the agent is moving in that period) and contemporaneous housing.

Like Cocco, I use equally-spaced grids across the two continuous state variables. Due to the presence of debt, I allow for negative cash-in-hand values.

Unlike Cocco, which used cubic-spline interpolation, I used linear interpolation over cash-in-hand (so I treated house size as a discrete choice with multiple levels - bilinear interpolation over cash in hand and housing led to little difference). As the investor must make two discrete choices each period (to move and to enter the stock market), their value function will be kinked. In this case, cubic spline interpolation methods may perform poorly (Van Hemert, 2010). Furthermore, cubic-spline interpolation methods are vulnerable to “overshooting” if regions with significant value function curvature are admitted into the state space (Guvenen, 2023). As the possibility of ending up near a constraint in the future (e.g. the positive consumption constraint) likely significantly influences behavior today, it is important to capture these regions. A choice which leads to potential default (or lands cash on hand below the grid) in the next period received a very low punishment utility.

A summary of my choice of grids:

- Housing: 10 grid points between \$20000 and \$250000
- Cash: 101 grid points between $-\$150000$ and \$2.5 million.
- Stock Share: 10 grid points between 0.0 and 1.0.
- Loan-to-value ratio: 15 grid points between 0.0 and $0.85 = 1 - d$.
- Consumption Share: 51 grid points between 0.001 and 1.0.
- Aggregate State: three grid points chosen using (Tauchen and Hussey, 1991).
- Already entered the stock market and Involuntary Movement grids are binary.

My initial house price index started at 1.0.

Discretizing transitory and persistent income:

Like Cocco, I use the (Tauchen and Hussey, 1991) method to approximate the continuous random processes each investor faces.

Noticing that all random variables (stock market return shocks, transitory shocks and aggregate shocks) are uncorrelated, I discretized them separately. House prices, as they are perfectly correlated with the aggregate state, do not need to be separately discretized and instead house-price grid points are obtain via $\frac{1}{\kappa_\eta}\eta$, where κ_η is simply the ratio of the variance of the aggregate state over the variance of the persistent component.

Like Cocco, I used 3 grid points for the aggregate state. I also used 3 grid points for both the transitory and stock market returns shocks (increasing the number of these shocks reduced participation by a small amount at great computational cost).

The distribution of the initial persistent component, η_0 , is taken from the stationary distribution of η_t found using Monte-Carlo Simulation.

⁹In Appendix F I go into the details of how I have implemented the standard backwards induction solution

Simulation

I then simulate lifecycles for each cohort and record the values each household attains of each relevant variable in each year. I simulate a number of households such that the proportion of observations by age / education group matches the PSID proportion presented in the paper.

Notes on the simulation:

- Replication Table 1 is constructed allowing both aggregate and transitory states to vary across lifecycles. However, the initial aggregate state individuals enter with must be specified. The initial state in my simulation is drawn from the stationary distribution of the aggregate state.
- Due to the discrete choices individuals face in my model (E.g. Stock market entry & whether to move), interpolating the policy functions for the simulation can result in fractional choices. To avoid this issue, (only) in my simulation I rounded off-grid values for cash-in-hand to their nearest grid point. This follows the simulation code of, for instance, (Campbell et al., 2021).
- A standard simulation exercise will generate a panel where all ages are weighted equally. To match the age proportions reported in the paper, for each education - age group, I generate a number of lifecycles in correct proportion to the total number of simulations. For each lifecycle, I retain only their choices at a single age. So my dataset is really a stacked cross-section rather than a panel.

C: Five-Year Scaling

Table 1. Five-Year Parameters used for the replication

Description	Parameter	Scaling	Description	Parameter	Scaling
Risk aversion	γ	Unscaled	Expected house price growth	b	$5b$
Discount factor	β	β^5	SD of house prices	$\sigma_{\hat{p}}$	Unscaled
Preference for housing	θ	Unscaled	Aggregate persistence	ϕ	ϕ^5
Down payment	d	Unscaled	<i>SD idiosyncratic income shocks</i>		
Depreciation rate	δ	$1 - (1 - \delta)^5$	No high-school degree	$\sigma_{\omega,1}$	Unscaled
Involuntary move	π	$1 - (1 - \pi)^5$	High-school degree	$\sigma_{\omega,2}$	Unscaled
Transaction cost	λ	Unscaled	College	$\sigma_{\omega,3}$	Unscaled
Gross Riskless rate	$\hat{R}_F$	$\hat{R}_F^5$	SD persistent income shocks	σ_{η}	Unscaled
Gross Mortgage rate	R_D	R_D^5			
Expected log stock return	μ	5μ			
Std. of log stock return	σ_{η}	$\sqrt{5}\sigma_{\eta}$			
One-time entry cost	F	Unscaled			
Minimum House Size	H_{min}	Unscaled			

(a) Preferences and stock returns

(b) Housing and income-process parameters

Table 2: Parameters used in the model

Notes: Agents are restricted to choose housing once every five years. It is also assumed income shocks only hit once every five years. Alternative scalings for these processes are summarized in Appendix E.

D: Age-Education Weights

Table 3: Age-education distribution

Age group	No highschool	Highschool	College	Total
25–29	0.01	0.06	0.02	0.09
30–34	0.02	0.08	0.04	0.14
35–39	0.01	0.10	0.06	0.17
40–44	0.02	0.08	0.07	0.16
45–49	0.01	0.05	0.03	0.09
50–54	0.02	0.04	0.02	0.07
55–59	0.02	0.04	0.02	0.08
60–64	0.02	0.03	0.02	0.08
65–69	0.02	0.04	0.01	0.07
70–74	0.02	0.03	0.01	0.06
Total	0.17	0.54	0.29	1.00

Notes: Each cell reports the share of the total sample in the indicated age-education cell; row totals equal the final column.

Taken from (Cocco, 2005).

E: Robustness

In this section, I outline several alternative calibration approaches for the labor and house price parameters I considered and why I did not use them for the main results. None of these approaches noticeably reduced the gap between my model-predicted stock-market participation rates and those reported in the paper.

Alternative approach	Alternative value	Rationale	Reason not selected
Scaling labour-income process parameters to five-year values (sum of random variables)	• $\text{Var}(\eta_5) = 13.6 \times \text{Var}(\eta_1)$ • $\text{Var}(p_5) = 4.1 \times \text{Var}(p_1)$ • $\text{Var}(\omega_5) = 4.1 \times \text{Var}(\omega_1)$ • $\phi_5 = 0.415$	If Cocco ran his regressions at the <i>annual</i> level in the PSID (as suggested by my empirical section), this applies the same method that turns the deterministic income profile into random shocks.	1. Running the Cocco method on <i>five-year</i> discounted household income recovers five-year persistent and transitory variances directly. These are close to the annual estimates, not many multiples of them. 2. Because the aggregate and house-price states are perfectly correlated, their variances would also have to be scaled. This inflates house-price risk and prevents the grid from matching PSID residuals (needed for Replicated Table 2).
Not scaling labour-income parameters at all	• $\text{Var}(\eta_5) = 0.019^2$ • $\text{Var}(\omega_5) = \text{Var}(\omega_{i,t})$ • $\phi_5 = 0.75$	If Cocco's regressions were actually run at the <i>five-year</i> level, no scaling of labour-income parameters is necessary.	1. Housing-process scaling would still break perfect correlation because the housing parameters are annual. 2. Five-year persistence of 0.75 for the aggregate labour-market/housing state seems high (my PSID estimate was 0.41). 3. Fails to match Cocco's deterministic lifecycle profile closely. 4. The variance of the persistent process closely matches the annual variance estimate of the permanent process presented in Cocco's (1999) thesis (section 4), using an identical method of estimation on an identical time period of the PSID and labor income construction description.
Starting value of the house-price index	$P_{\text{init}} = 0.7$	Matches the ratio of 1970 to 1992 house-price levels.	Essentially lowers the relative price of houses, so stock-market participation jumps even earlier because the initial cost of purchasing housing falls.

Table 4: Alternative calibration choices and reasons for rejection

F: Detailed Description of Solution

For each state, I looped over equally spaced grids over choices as described in Appendix B.

Given a chosen loan to home value ratio, we can use that the house price level is pinned down by the period (1 to T) and the value of the persistent state. Therefore, a choice of LTV ratio and housing implies a unique choice of debt given the house price is known and has a convenient space (0 to 0.85).

Similarly, maximum consumption levels were found by checking whether the BOTH contemporaneous house size choice in that particular loop equals the current housing level AND that the involuntary movement shock is set to zero. If so, this is the level of consumption that sets stocks and bonds to zero in the “No Move” budget constraint (given the other choice/state combination of that loop). If not, I perform the same operation on the “Move” budget. The consumption choice is then the consumption share times the maximum feasible consumption.

After a unique debt, consumption, housing, stock market entry and stock share choice are determined, I impose the following conditions BEFORE I compute continuation values:

1. Individuals cannot have positive stock share if they are not in the stock market.
2. If the agent has already entered the stock market, they will not do so again.
3. If the maximum consumption level or stocks and bonds are negative, skip.
4. If in the first period, skip if the housing state is not 0.0 (as individuals start with zero housing).
5. If **not** in the first period, skip if the housing state is 0.0 (which would violate the minimum housing constraint).
6. If in the stock market, skip if the LTV ratio is positive **and** the stock share is less than 100%. As debt and bonds are both risk-free, agents will not hold them simultaneously. (Technically, this applies both when in the stock market and not, but the nature of grid-search over choices may imply that small amounts must be allocated to bonds if not in the stock market - and I don't want to force investors to enter because they cannot exhaust their budget otherwise).

Relaxing this constraint does not affect the final policy but slows down the solution.

If no choice is feasible for a particular state, then I assigned punishment utility to that state.

Computing Continuation Values

After computing whether an individual is moving or not and their choice, I use the budget constraint and the grids over random variables to compute their possible continuation values.

I use α and the budget constraint to find stocks (S) and bonds (B). For a retiree, the level of labor income tomorrow (Y') is known. For a worker who will not retire next period, it will vary with the persistent and transitory state. (I show the retiree computation below, but the worker's computation follows with this adjustment).

Then, the value of the choice combination is found using:

$$\text{val} = U(c, H') + \beta \sum_{\iota'} \sum_{\eta'} T_{\eta}[\eta, \eta'] \cdot T_{\iota}[1, \iota'] \cdot [(1 - \pi_m) \cdot v^{\text{no move}}(X') + \pi_m \cdot v^{\text{move}}(X')]$$

where:

$$\begin{aligned} R' &= \exp(\iota' + \mu) \\ X' &= R' \cdot S + R_F \cdot B - R_D \cdot D + Y' \\ Y'_i &= Z_{i,t+1} \end{aligned}$$

and:

- β : discount factor
- π_m : probability of being forced to move
- $T_{\eta}[\eta, \eta']$: transition probability from current income shock η to future shock η'
- $T_{\iota}[1, \iota']$: transition probability from current return shock to future return shock ι' (I.I.D.)
- $v^{\text{no move}}(X')$, $v^{\text{move}}(X')$: interpolated value functions at future cash-on-hand X' , conditional on not moving or moving, respectively. For each $(\eta', \text{Stock Market Entry Tomorrow, Housing Tomorrow, Involuntary Move Tomorrow})$ combination, there is a different value function. Bilinearly interpolating over housing did not change the results.

References

- Campbell, J. Y., Clara, N., and Cocco, J. F. (2021). Structuring mortgages for macroeconomic stability. *The Journal of Finance*, 76(5):2525–2576.
- Cocco, J. F. (2005). Portfolio choice in the presence of housing. *The Review of Financial Studies*, 18(2):535–567.
- Cocco, J. F., Gomes, F. J., and Maenhout, P. J. (2005). Consumption and portfolio choice over the life cycle. *The Review of Financial Studies*, 18(2):491–533.
- Fonseca, J., Liu, L., and Mabilie, P. (2024). Unlocking mortgage lock-in: Evidence from a spatial housing ladder model. Working paper.
- Güvenen, F. (2023). Lecture 2: Function interpolation. https://static1.squarespace.com/static/6246570e617f1d3daf55e1c1/t/654d240e20de9223fc1c9492/1699554319026/Lecture2_INTERPOLATION_2023_WEB.pdf. Lecture slides, *PhD Computational Methods* course, University of Minnesota.
- Tauchen, G. and Hussey, R. (1991). Quadrature-based methods for obtaining approximate solutions to nonlinear asset pricing models. *Econometrica*, 59(2):371–396.
- Van Hemert, O. (2010). Household interest rate risk management. *Real Estate Economics*, 38(3):467–505.